1. **Aim of the practical:** Implement the validations in forms using jQuery.
2. **Objective:** To enhance user experience and ensure data integrity by validating form inputs dynamically using jQuery before submitting the form to the server.

3. **Commands/Steps:**

Index.html

```
  <!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Login Form</title>
   <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
</head>
<body>
   <div class="container">
      <h2>Information Form</h2>
      <form id="loginForm">
         <label for="email">Email:</label>
         <input type="email" id="email" name="email">
         <span class="error" id="emailError"></span>
```

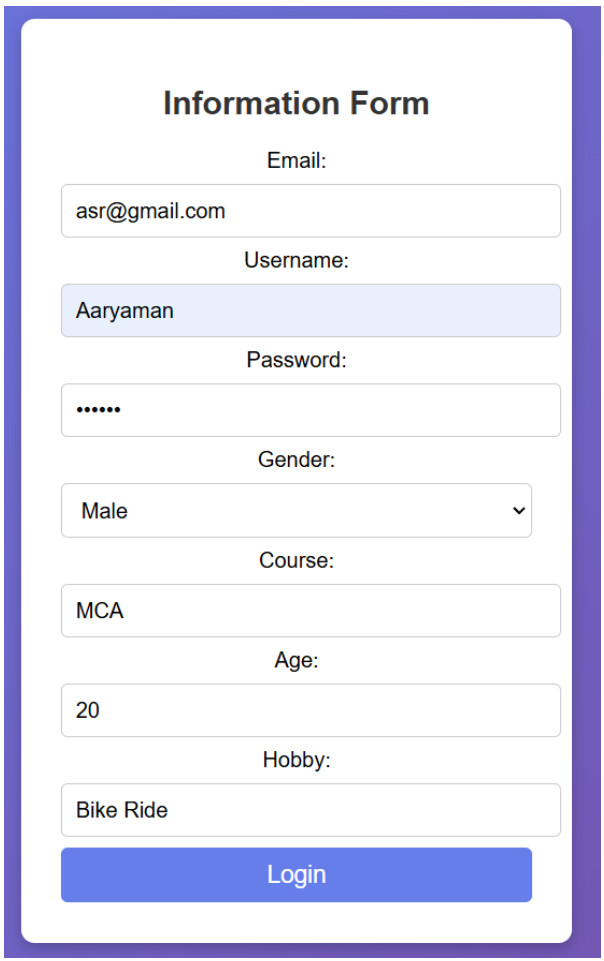
```
        <label for="username">Username:</label>
        <input type="text" id="username" name="username">
        <span class="error" id="usernameError"></span>

        <label for="password">Password:</label>
        <input type="password" id="password" name="password">
        <span class="error" id="passwordError"></span>

        <label for="gender">Gender:</label>
        <select id="gender" name="gender">
            <option value="">Select Gender</option>
            <option value="male">Male</option>
            <option value="female">Female</option>
            <option value="other">Other</option>
        </select>
        <span class="error" id="genderError"></span>

        <label for="graduation">Course:</label>
        <input type="text" id="graduation" name="graduation">
        <span class="error" id="graduationError"></span>

        <label for="age">Age:</label>
        <input type="number" id="age" name="age" min="1">
        <span class="error" id="ageError"></span>

        <label for="hobby">Hobby:</label>
        <input type="text" id="hobby" name="hobby" min="1">
        <span class="error" id="hobbyError"></span>

        <button type="submit">Login</button>
    </form>
</div>
```

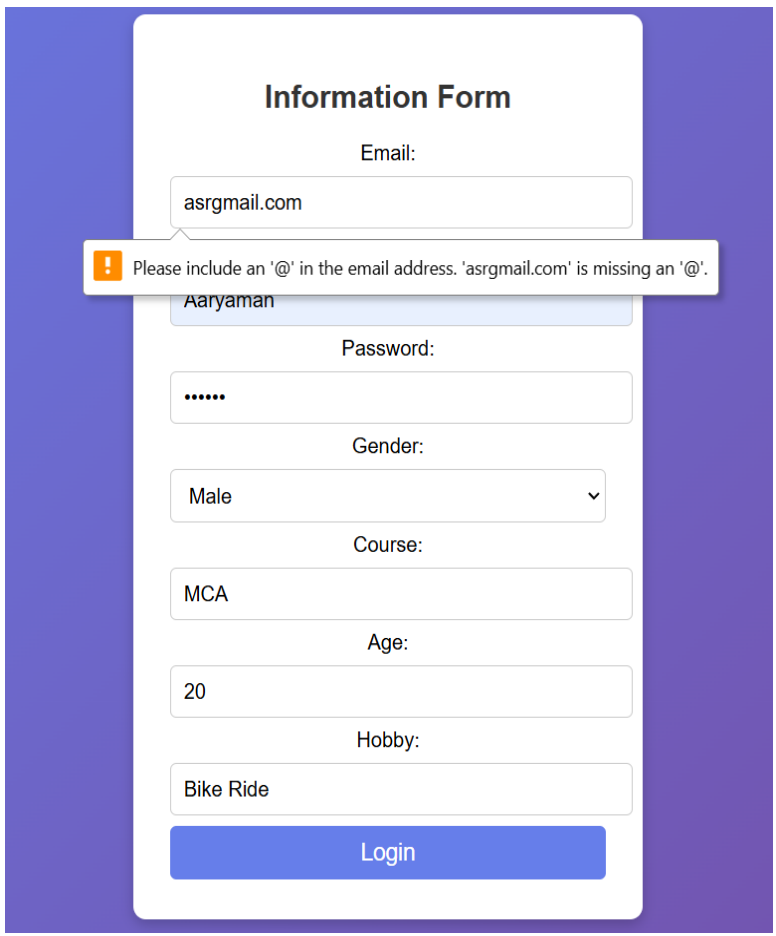
JavaScript

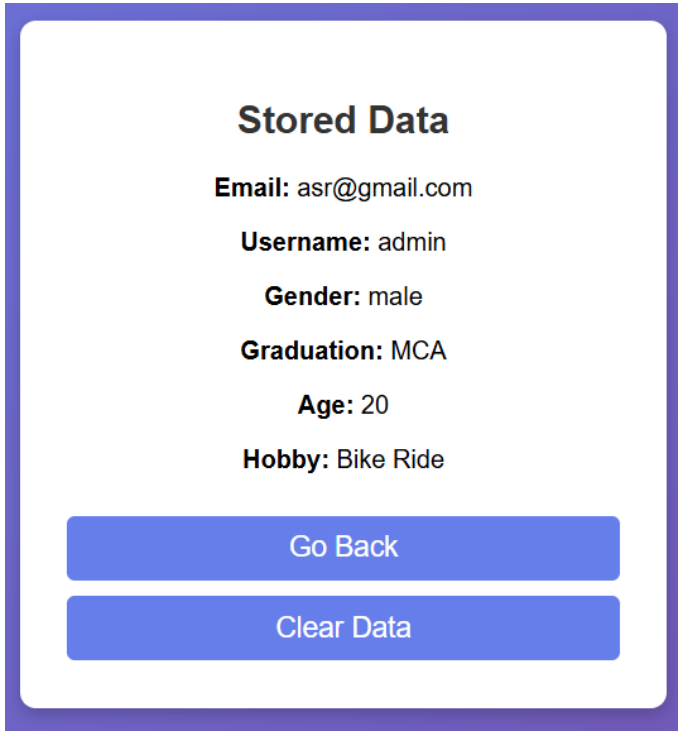
```
<script>
        $(document).ready(function() {
    $("#loginForm").submit(function(event) {
        let isValid = true;
        $(".error").hide();
        $("#successMessage").hide();

        let email = $("#email").val().trim();
        let username = $("#username").val().trim().toLowerCase().replace(/\s/g, "");
        let password = $("#password").val().trim();
        let gender = $("#gender").val();
        let graduation = $("#graduation").val().trim();
        let age = $("#age").val().trim();
        let hobby = $("#hobby").val().trim();

        $("#username").val(username);

        email === "" && $("#emailError").text("Email is required").show() && (isValid = false);
        !/^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(email) && email !== "" &&
$("#emailError").text("Enter a valid email").show() && (isValid = false);
        username === "" && $("#usernameError").text("Username is required").show() &&
(isValid = false);
        password === "" && $("#passwordError").text("Password is required").show() && (isValid
= false);
        password.length < 6 && password !== "" && $("#passwordError").text("Password must be
at least 6 characters").show() && (isValid = false);
        gender === "" && $("#genderError").text("Please select gender").show() && (isValid =
false);
        graduation === "" && $("#graduationError").text("Graduation is required").show() &&
(isValid = false);
        (age === "" || age <= 0) && $("#ageError").text("Enter a valid age").show() && (isValid =
false);
```

```
        (hobby === "" || /\d/.test(hobby)) && $("#hobbyError").text("Enter a valid hobby (no numbers)").show() && (isValid = false);

        !isValid && event.preventDefault();
    });
});
```

4. Result:







5. Learning Outcomes (what I have learned):

- Users receive instant feedback on input errors, reducing frustration and improving form submission accuracy.
- Prevents incorrect or incomplete data from being submitted to the server, ensuring valid and structured input.
- Reduces server load by handling validation on the client side, leading to faster processing and better performance.